COMP9120 / DATABASE MANAGEMENT SYSTEMS

WORKING PROTOTYPE · SYNTHETIC DATA

Distance should not decide access.

Remote Education Resource Support System

A connected workflow for student needs, school review and accountable resource allocation.

01 / THE CHALLENGE

Support is fragmented.

Remote learners face uneven device and internet access. Schools need a consistent way to connect needs, verified eligibility and available resources.

02 / OUR RESPONSE

One traceable journey.

A Flask interface connects the team's Python services to PostgreSQL: discover support, submit a learning goal, review, allocate, record usage and report.

03 / DATABASE VALUE

Consistency at the core.

Foreign keys, checks, unique constraints and transactional row locks protect relationships and prevent duplicate or excessive allocations.

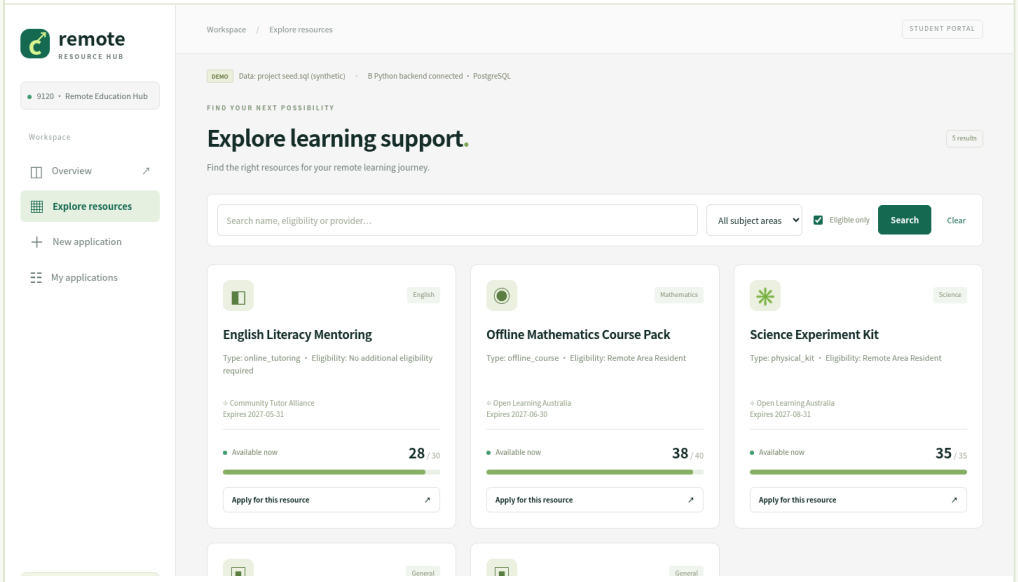
From need to measurable support

A REAL DATABASE WORKFLOW

01 **Discover eligible resources** → 02 **Submit a learning goal** → 03 **School-scoped review** → 04 **Approve + allocate atomically** → 05 **Track usage and report**

Student / resource discovery

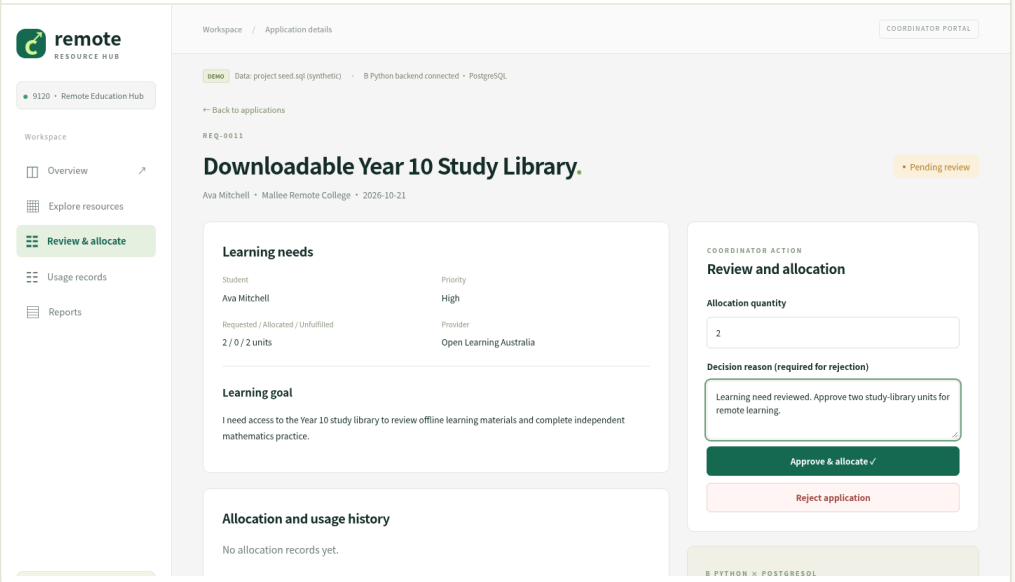
LIVE UI CAPTURE



Verified eligibility, expiry and remaining quantity guide discovery. A pending application does not reserve capacity.

Coordinator / review and allocation

LIVE UI CAPTURE



Only the student's school coordinator can review. Approval and allocation commit together, or both roll back.

IMPLEMENTED ARCHITECTURE

Flask interface

Student, coordinator, usage and report pages

`CSRF · role + school scope`

→

Team B / Python

submit_application

approve_and_allocate

`reject_application · add_usage_record`

→

PostgreSQL

14 related tables · 2 views

Eligibility + capacity checks

`FOR UPDATE · COMMIT / ROLLBACK`

Key relationships: Region → School → Student → Application → Allocation → Usage record. Eligibility and funding use many-to-many relationships; provider and funder are overlapping organisation subtypes.

VERIFIED, NOT JUST WIREFRAMED

47 passing integration checks against real PostgreSQL

- Application → approval → allocation → usage → CSV
- Capacity, expiry and eligibility failures roll back
- Cross-school access and repeat approvals blocked
- Concurrent requests cannot take the same last unit
- Desktop workflow + 390 px mobile layout verified

PROJECT SEED DATA

Designed for a demonstrable system.

5	10	8	14
REGIONS	STUDENTS	OFFERINGS	TABLES

Scope: synthetic seed data and a password-protected demo account. Production identity integration is future work. Initial partial allocation is supported; post-approval top-ups and automatic stock return are not exposed by the supplied backend.

Deliverables: runnable source, SQL, integration evidence, one-page poster, captioned walkthrough and a live-demo runbook.

Source: project schema.sql, seed.sql, Python services.py and queries.py.
No real student data. Results describe this prototype, not measured educational impact.